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Purpose/Policy:

“This changeover guide will be used to instruct a user to change the “Cartoner” over to 12oz 4pk units from any size. This guide will include multiple changeover formats into the 12oz 4pk format and will also include basic troubleshooting guides, cleaning guides and operations.”

Scope:

"The Cartoner is responsible for package cans into wraps via a continuous motion drive chain and bucket chain."

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- CIP – Cleaning in Process
- UHMW – Ultra-High Molecular Weight plastic (used for guides and wear strips)
- Cartoner – Machine that packages cans into wraps (cartons)
- Switchback Tray Packer – Machine that puts wraps, Paktech units or loose cans into trays, or passes wrapped cases through

Authorities/Responsibilities:

"This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to assist with "12oz 4pk Cartoner Changeovers". A trainer will be responsible for show how to make the changeover adjustments in correlation with this SOP document."

Resources and Materials:

Operations:

- Air.
- Wraps.
- Full Product Cans.
- Chemlube-10.

Changeovers:

- 4pk/6pk Buckets.
- 4pk/6pk UHMW Flight Block.
- 4pk/6pk Pusher arm Block.
- Universal Flight Adjustment Tool.

Tools:

- Pneumatic Torque Wrench (for Bucket removal).
- ½ Wrench (or 13mm).
- 7/16” Wrench.
- 9/16” Socket.
- 5mm Hex Wrench or 3/16”.
- Square.

Basic Cleaning:

- Alpha 5.
- Warm Water.
- Dry Rags.
- Shop Towels.
- Dawn Soap.
- Glue Scraper.

Deep Clean:

- Refer to Zone Cleaning SOP.
    - N:\Operations\Production\Boston\Packaging\Canning Line 2024\Zone Cleaning\Can Line Lower Level\Zone 7 - Lower Level - Cartoner, Tray Packer

Troubleshooting:

- No Equipment required.

PPE Required:

- No formal PPE required, suggested nitrile gloves for cleaning.

Procedures:

Changeover Guide:

Recipe Data:

1. From the HMI screen, select “Recipe Data.”
2. Using the arrows, select “Recipe #1 4 PAC 12OZ.”
3. *Select “Setup Data.”*

Adjusting the Magazine Deck:

1. Using the HMI, reference #20 – “Mag Back Guide” to start setting up the Magazine Deck.

2. Reference #1 to set up the “Mag Front Guide.”
3. Reference #2 to set up the “Mag Top Guide.”
4. Reference #3 to set up the "Mag Deck”

Adjusting the Top Guides:

1. Reference #15 for “Top Guide Long Left” adjustment.

Note: For more detailed set-up, refer to operator training.

2. Reference #16 for “Top Guide Left” adjustment.
3. Reference #17 for “Top Guide Center” adjustment.
4. Reference #18 for “Top Guide Right” adjustment.
5. Reference #19 for “Top Guide Long Right” adjustment.

Note: Make sure that the “Top Guide Long Right” is not being hit by pick arm.

Note: For more detailed set-up, refer to operator training.

Adjusting the Erector Height:

Note: This adjustment will only be needed when packaging different can sizes (16oz or 19.2oz).

1. *Loosen the red knobs located on each side of the “Erector Housing.”*
2. *Reference #4 on the HMI for the “Erector Housing” height.*
3. *Tighten the red knobs that were loosened in step #1.*

Adjusting the Nose Bars:

1. Reference #12 on the HMI for the “Load Side Nose Bar” adjustment.
2. Reference #13 on the HMI for the “Non-Load Side Nose Bar” adjustment.
3. Reference #14 on the HMI for the “Nosebar stick out” adjustment. Please note there are two “Nosebar stick outs.”
4. Ensure the orange fasters for the “Nosebar stick outs” are not in the way of the cartons.

Overhead Rails (Raising):

1. Raise the overhead rails using the actuated lever switch located next to the light screen curtain.
2. Reference #6 “Overhead Upstream” and #7 “Overhead Downstream” for adjustments.
3. If changing from 16oz or 19.2oz products, reference #8 “Overhead Vertical” to adjust the Overhead Rails down.

Note: At this point, do not lower the "Overhead Rails," this step will be completed later in the document.

Adjustments if coming from 19.2oz:

1. Lower load side major flap guide rail.
2. Lower Wheels on compression entrance.
3. Lower operator side major flap guides (first down by the flap tucker wheel and second the upper guide rail).

Adjusting the Vacuum Arms:

1. Reference #21 "Pick Vac Shutoff" for the vacuums that need to be disabled.
2. Reference #28 "Bear Claw Down" to determine if the claws should be in the down position. If it says no, ensure they are in the up position.

Infeed Lanes:

1. When changing packaging sizes, the infeed lanes need to be opened (for more cans) or closed (for less cans).
2. Reference #29 "Infeed Lanes Used" to determine how many lanes are needed.
3. Loosen the two knobs on the "Infeed Lane Plates."
4. Adjust the plate to the number of lanes needed.
5. Tighten down the two knobs on the "Infeed Lane Plate."
6. If changing over from 16oz or 19.2oz cans, set the recirculation sensor per the mark on the sensor.

Glue Position:

1. Reference #24 "Load Side Glue Vert" and #25 "Non-Load Glue" for the glue gun height adjustment.

Note: Refrain from adjusting any other parts of glue guns unless maintenance is present.

2. Loosen the orange handle and raise/lower the glue guns until the height required is reached.
3. If coming from 19.2oz, lower glue sensor.

Center Support:

1. Depending on the size of the previous run, the center support may need to be removed.
2. Reference #30 "Center Support Used" for instructions.

Adjusting Flights:

Note: If changing over from 12oz 6pks, the flights will not need to be adjusted.

1. *At the outfeed of the Cartoner, loosen the four "Flight Chain Locking Screws" using a 9/16" Socket.*
2. *Place the 4pk/6pk UHMW Flight Block in between the flights.*
3. *Using the* Universal Flight Adjustment Tool, close the flights till the lead flight is almost snug with the *4pk/6pk UHMW Flight Block.*
4. *Once the flights are set, tighten down the four "Flight Chain Locking Screws" using a 9/16" Socket.*

Pusher Arm Block:

1. Using a 5mm Hex Key and one 3/16" bolt, add the 4pk/6pk Pusher Arm Block along with cardboard shim.

Adjusting the Strongback:

1. Loosen the orange lock handle above the glue compression pads at the outfeed of the Cartoner.

Note: Ensure the center bar is removed before adjusting the "Strongback" if present.

2. Reference #5 "Strongback" on the HMI for setting number.

Note: If any counter seems like it is not reading correctly, contact maintenance. In lieu of the counter, the square tool can be used to measure the width of the "Strongback" and can be used to set the size correctly.

3. *Once completed, tighten down the orange lock handle that was loosened in step #1.*

Overhead Rails (Lowering):

1. Using the actuated lever switch located next to the light screen curtain, lower the "Overhead Rails."

Swapping out buckets:

Note: 4pk Wraps and 6pk Wraps use the same set of buckets. If changing from 12oz 4pk to 12oz 6pk, this section can be skipped. If changing from 12pks or from15pks, please follow the instructions below.

1. *From the HMI, select the "Main" button to get to the home screen.*
2. *Place the Cartoner into "Test" mode by pressing the "Test" button.*
3. *Using the Pneumatic Torque Wrench, remove the bolt that holds the bucket in place.*
4. *Ensure the buckets are cleaned.*
5. *Replace the bucket with the 4/6pk bucket and tighten down the bolt.*
6. *Jog the machine one bucket at a time until all buckets have been replaced.*

Loading the Recipe:

1. After all the above sections have been completed, the recipe will now need to be loaded.
2. From the HMI panel, press the "Reset" button.
3. Ensure the Cartoner is still in "Test" mode.
4. Using the HMI, select "Setup Data" and select "Use" on Recipe #1.

Setting up and Operating the Cartoner:

1. From the home screen, press the "Run Mode" button to engage packaging operations.
2. Turn on the Glue by pressing the power button twice on the Glue Unit HMI if the screen is not lit. If the screen is lit, press the power button once. Flashing green indicates that the unit is warming up and orange is in standby mode.
3. Load the packaging material needed for the current packaging run into the staging area.
4. Load the Cartoner Magazine with the current wraps needed for the run.
5. From the HMI home screen, turn on the following:
    5.1 "Glue."
    5.2 "Vacuum Pick."
    5.3 "Magazine Conv. On."
6. From the HMI, select "Packer."
7. From the Packer Screen, select "Prime Mode" and then select the green "start" button from the HMI panel.
    7.1 This will prime the infeed lanes with cans.
8. Once the machine is "Primed" with cans, deactivate "Prime Mode" and press the "Main" button on the HMI.
9. From the HMI panel, press the green start button to start packaging cases.
10. Inspect cases for uniformity, glue shot placement and structure. Adjust machine as needed.

Note: For basic adjustments and troubleshooting, please refer to the "Troubleshooting Guide" section of this SOP document.

Shutting down the Cartoner:

1. After the “push” has been completed by the Can Line Filler Operator, the Cartoner will need to be “Packed out.”

Note: Please see below for instructions and refer to operator training on how to complete.

2. From the HMI, select “Packer.”
3. Select “Purge On.”
4. Press the “Start” button from the Cartoner HMI Panel.
5. Let 4-5 wraps be picked by the pick arms and then select “Vacuum Pick” to turn off the Vacuum.
6. Ensure all cans are removed from Cartoner before next run or before cleaning.

Basic Cleaning of the Cartoner:

Note: A hose should never be used inside the Cartoner and please note the water should never be sprayed near the control panels or HMI screens.

1. CIP Rinse:
    1.1 Under the Cartoner HMI panel there are two levers for the CIP rinse and air blow off.
    1.2 While the machine is in runout mode and running, open and hold the CIP rinse lever until all flights and chains are rinsed thoroughly.
    1.3 After the Cartoner has been rinsed, use the Air lever to air dry the Cartoner flights and chains.
2. Using a spray bottle of “Alpha 5”, spray the cartoner stainless steel areas and buckets.
    2.1 Wipe areas with a dry rag until clear of beer build-up and dust accumulation.
3. Using a spray bottle of “Alpha 5”, spray the pusher arm areas above and below. This area can be a cause of frequent case crashes. Ensure this area is cleaned.
4. Use a rag and “Alpha 5” to clean the plexiglass windows. Never use a scrub pad on plexiglass.
5. Use the air gun to blow out any cardboard dust present in or on the machine.

Deep Cleaning:

1. Refer to Zone Cleaning SOP.
    1.1 N:\Operations\Production\Boston\Packaging\Canning Line 2024\Zone Cleaning\Can Line Lower Level\Zone 7 - Lower Level - Cartoner, Tray Packer

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

Note: Listed below are common issues that may happen during a packaging run. This SOP contains basic troubleshooting techniques to help resolve some of these issues. If the issues cannot be resolved with the help of this SOP, please reach out to a senior member of the can line team or Maintenance team.

1. Wraps failed to be picked:

Note: By using "Test Mode" it is easier to help diagnose the issues below.

- 1.1 Ensure the wraps are flush in the pick area and not bowed.
- 1.2 Ensure the magazine is not squishing the wraps.
- 1.3 Ensure the correct suction cups are on as stated in this SOP Document.
- 1.4 Ensure the "Top Guide Long Left" and "Top Guide Long Right" are not too tight.

2. Wraps not being placed into pocket:

Note: By using "Test Mode" it is easier to help diagnose the issues below.

- 2.1 Ensure the "Bear Claws" are set as stated in this SOP document.
- 2.2 Verify the flight spacing.
- 2.3 Verify the "Nosebar" is in the correct position.
- 2.4 Inspect the "Prebreak" area.
- 2.5 Verify the "Erector" is in the correct position.

3. Minor Flaps not tucking:

Note: By using "Test Mode" it is easier to help diagnose the issues below.

- 3.1 Verify the "Strongback" is set correctly as outlined in this SOP document.
- 3.2 Ensure the railings are clean and have not been moved.
- 3.3 Inspect wrap for stiff cardboard.
- 3.4 Verify the Magazine is set as stated in this SOP document.

4. Cans tipping over in infeed lanes:
   - 4.1 Verify the infeed of the Cartoner has an adequate amount of line lube.
   - 4.2 Verify the "Recirculation Sensor" is at the correct height.
   - 4.3 Verify there are no down cans in the recirculation table of the Cartoner.
   - 4.4 Inspect the infeed lanes for tackiness and clean as needed.
   - 4.5 Inspect infeed lanes for down cans.

4.6 Inspect conveyor belt for any chips/broken pieces.
4.7 Inspect for awkward transfer between stages.
4.8 Ensure major flap is tucker on load side of the machine.

5. Cans rolling out of package:
   5.1 Verify the “Pusher Arm” is pushing all the cans into the wrap.
   5.2 If the wrap is “popped” up, cans will not transition smoothly causing a down can which results in a roll out.
   5.3 Verify the flights are within 1/8” the buckets with black tooling and there is no backlash present.
      5.3.1 If the buckets seem out of time, please contact the maintenance department.
6. Removing bad wrap:
   6.1 Stop the Cartoner.
   6.2 Remove trouble wrap.
   6.3 From the HMI, select “Troubleshooting.”
   6.4 Press the “White Square” where the wrap would be in the machine.
   6.5 Restart the Cartoner.
7. Glue shot formation:
   7.1 Inspect the glue shot of the wrap to ensure glue is being applied to the carton and not the cans.
   7.2 With assistance from a senior member of the can line team or maintenance, adjust the glue gun position as needed to achieve proper glue shot.
8. Wrap is skewing (not square):
   8.1 Verify the “Strongback” is set correctly as stated in this SOP document.

Note: Always check the “Strongback” prior to checking the compression pads.

   8.2 Inspect the “Compression Pads” for glue build up and remove any build up with glue scraper.
   8.3 Ensure the bolts on the compression bads are tight and not sticking through.
   8.4 Inspect the “Compression Pads” to see if there is enough pressure being placed against the wrap. Adjust as needed.
   8.5 Inspect the rails as the wrap transfers to the “Compression Pads.”
9. Wrap non-existent machine at the Cartoner outfeed:
   9.1 Inspect the “Strongback” setting as stated in this SOP document.
10. Cartons not filling:
   10.1 Consistent flight issue – Ensure skid pads are not sticking up off the flight chain.
   10.2 All Flight/Most flights – Ensure both carton sensors are flagging fully without stuttering while carton is placed. If eye is not flagging or stuttering, grab maintenance.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
001	2024/05/28		Kyle Szymanski

002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

Version Approval Number: 002

Approved by: Kyle Szymanski

Position Title: OSCI Manager

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